Assignmet Transcription Image to LaTeX conversion

11/08/2026 21:26:53
https://claude.ai/chat/e5f85b59-a1d8-405c-be56-c487d04eb9ef

Take this image and write these in latex following /iee

I've turned your three columns into one vertical `gather*` block in your IEEE...

no need for the whole file.

```latex \begin{gather*} \underline{0 < t < DT_s} \text{ ($S_1$ on):}\\[1ex]...

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```latex \begin{gather*} \underline{DT_s \le t \le (D + D_2)T_s} \text{ ($S_1$ off,...

Take this image and write these in latex following /ieee-homework-latex no need f...

```latex \begin{gather*} \underline{\underline{\text{Core flux balance}}}\\ \qua...

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```latex \begin{figure}[!htbp] \centering \includegraphics[width=0.5\linewidth]...

I want you to optimize this

**The minimum conduction loss is at n = 0.56, where P_cond = 2.82 W.** That tur...

Transcribe this in latex

```latex \begin{gather*} \underline{\underline{\text{Input current}}}\\ \quad...

Transcribe this in latex

```latex \begin{gather*} \underline{\underline{\text{Duty ratio}}}\\ \quad V_o =...

I need to Simulate the same flyback three times (optimal n, 10% lower, 10% highe...

**The ±10% runs change the conduction loss by only 0.3–0.4%.** That's because...

show each calculations in detail so i can verify

Below are all three runs, worked step by step. Each uses the same seven formulas...

now give this in latex

I put the full write-up in `sweep.tex`. It's only the part to paste into your report, n...

I need a plecs initialization code for all three

Here are two files. `flyback_init.m` is the initialization code: change `RUN` to 1, 2 ...

Initialization commands of model 'flyback_converter' cannot be evaluated: operat...

**What's causing it?** `n` is ending up as a list of three values instead of one...

How to use the RMS block in Plecs standalone

**Put an ammeter in series with the MOSFET, feed its output into the RMS Value...

Create an answer to the problem 3 with this.

Now the core-loss chart, with the old 50 mT marks replaced by marks at the...

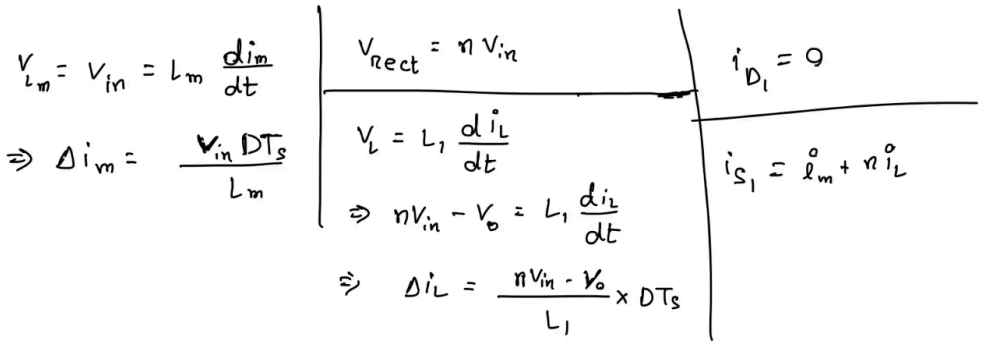
made some changes. Now I am to have two secondary winding for two load...

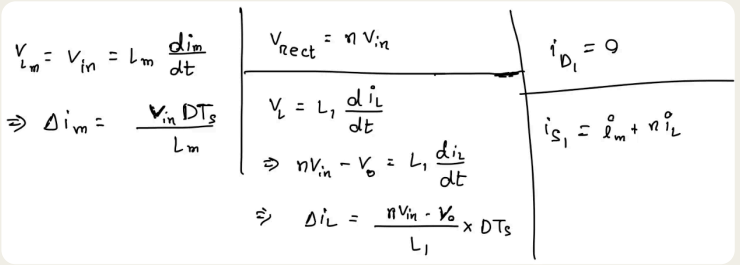
I wrote the two-output version of Problem 3 in `problem3_two_outputs.tex`, using...

Have the numerical calculations right next to each equation's derivations

Replacing (b)+(c) with one section where each formula is followed by its numbers,...

I rewrote your script with the new math in `problem3_final.tex`. The final design is...




Take this image and write these in latex following /iee

*11/1/2026, 12:32:06 PM*

- Loaded a skill
- Choosing the file format for the LaTeX output.
- Created a file
- Command failed
- Command failed
- Ran a command
- Ran a command
- Read a file
- Shared files
- Done

I've turned your three columns into one vertical `gather*` block in your IEEE homework template, saved as `forward_interval1.tex`. I compiled it with the IEEEtran class and it builds with no errors and no overfull lines.

**What's in it?** The three columns of your sketch are now three sub-headings under one time interval, each with its result boxed:

1. **Magnetizing inductance:** Δi_m = V_in·D·T_s / L_m
2. **Output inductor:** V_rect = nV_in, then L₁·di_L/dt = nV_in − V_o, so Δi_L = (nV_in − V_o)·D·T_s / L₁

3. **Device currents:** i_D1 = 0 and i_S1 = i_m + n·i_L

**Did the math check out?** Yes. All three are consistent if n is secondary turns over primary turns: the secondary gets nV_in, so the load current reflects back to the primary as n·i_L.

**What did I add that isn't in your image?**

- **Interval label "0 < t < DT_s (S₁ on)":** your sketch has no label, but it's the switch-on interval (you integrated over DT_s), so I added it. Change the wording if your circuit names the switch differently.
- **One lead-in sentence:** a short line before the equations. Delete it if you don't want it.

**What do you need to fill in?** `<COURSE CODE>`, `<COURSE NAME>`, `<ASSIGNMENT NAME>`, and `Problem N`. If you're pasting this into an existing file instead, copy just the block between the `% ===== Fragment` comment lines:

```
\begin{gather*}
\underline{0 < t < DT_s} \text{ (S_1 on):}\\[1ex]
\underline{\underline{\text{Magnetizing inductance}}}\\
\quad v_{L_m} = \Vin = L_m \frac{di_m}{dt}\\
\implies \boxed{\Delta i_m = \frac{\Vin D T_s}{L_m}}\\[1ex]
\underline{\underline{\text{Output inductor}}}\\
\quad \Vrect = n\Vin\\
\implies v_L = L_1 \frac{di_L}{dt}\\
\implies n\Vin - V_o = L_1 \frac{di_L}{dt}\\
\implies \boxed{\Delta i_L = \frac{n\Vin - V_o}{L_1}\, D T_s}\\[1ex]
\underline{\underline{\text{Device currents}}}\\
\quad i_{D_1} = 0\\
\implies \boxed{i_{S_1} = i_m + n\, i_L}
\end{gather*}
```

The block uses two macros, so put these in your preamble:

```
\newcommand{\Vin}{V_{\mathrm{in}}}
\newcommand{\Vrect}{V_{\mathrm{rect}}}
```

Also make sure the five compact-spacing `\setlength` lines come right after `{\small`. No images or bibliography are needed, so on Overleaf a single pdfLaTeX run is enough.

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no need for the whole file.

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```
\begin{gather*}
\underline{0 < t < DT_s} \text{ ($S_1$ on):}\\[1ex]
\underline{\underline{\text{Magnetizing inductance}}}\\
\quad v_{L_m} = V_{in} = L_m \frac{di_m}{dt}\\
\implies \boxed{\Delta i_m = \frac{V_{in} D T_s}{L_m}}\\[1ex]
\underline{\underline{\text{Output inductor}}}\\
\quad V_{rect} = n V_{in}\\
\implies v_L = L_1 \frac{di_L}{dt}\\
\implies n V_{in} - V_o = L_1 \frac{di_L}{dt}\\
\implies \boxed{\Delta i_L = \frac{n V_{in} - V_o}{L_1}\, D T_s}\\[1ex]
\underline{\underline{\text{Device currents}}}\\
\quad i_{D_1} = 0\\
\implies \boxed{i_{S_1} = i_m + n\, i_L}
\end{gather*}
```

This version doesn't need any macros in your preamble. It writes V_in and V_rect out directly, so you can paste it straight into your file.

The interval label "0 < t < DT_s (S_1 on)" is my addition, since your image doesn't have one. Change it if your switch has a different name.

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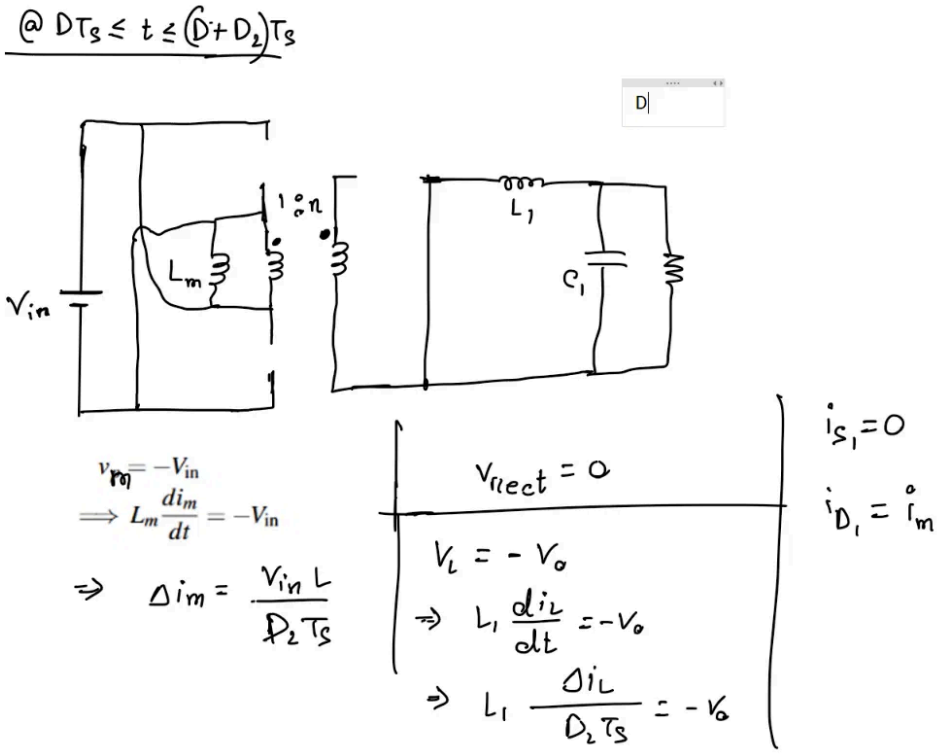

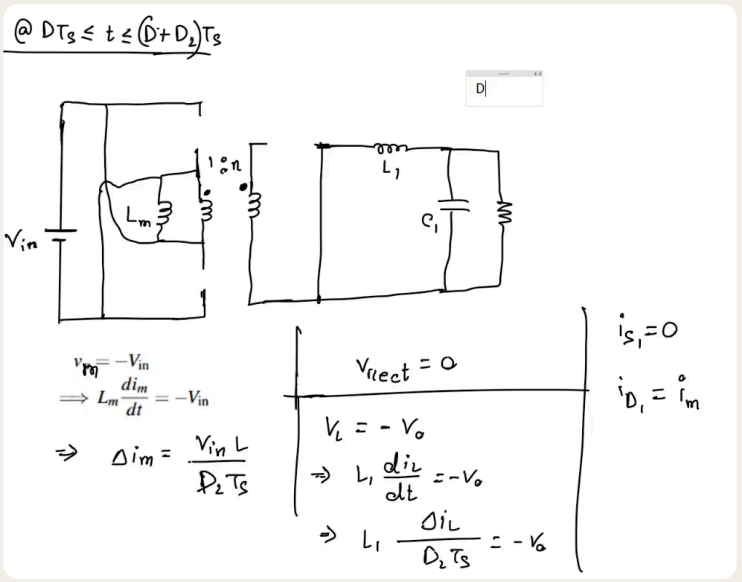

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Correcting the ripple expressions for the inductor.

Correcting the ripple expressions for the inductor.

Done

```
\begin{gather*}
\underline{DT_s \le t \le (D + D_2)T_s} \text{ ($S_1$ off, $D_1$ on):}\\[1ex]
\underline{\underline{\text{Magnetizing inductance}}}\\
\quad v_m = -V_{in}\\
\implies L_m \frac{di_m}{dt} = -V_{in}\\
\implies \boxed{\Delta i_m = -\frac{V_{in} D_2 T_s}{L_m}}\\[1ex]
\underline{\underline{\text{Output inductor}}}\\
\quad V_{rect} = 0\\
\implies v_L = -V_o\\
\implies L_1 \frac{di_L}{dt} = -V_o\\
\implies L_1 \frac{\Delta i_L}{D_2 T_s} = -V_o\\
\implies \boxed{\Delta i_L = -\frac{V_o D_2 T_s}{L_1}}\\[1ex]
\underline{\underline{\text{Device currents}}}\\
\quad i_{S_1} = 0\\
\implies \boxed{i_{D_1} = i_m}
\end{gather*}
```

Two changes from your sheet:

1. **Δi_m was wrong on your page.** You wrote Δi_m = V_in L / (D₂T_s), which has the inductance and the time interval swapped. Integrating L_m·di_m/dt = −V_in over D₂T_s gives Δi_m = −V_in·D₂T_s / L_m. The units confirm it: V·s/H gives amps, while your version gives V·H/s, which isn't a current.
2. **I solved the last line through to Δi_L.** Your page stops at L₁·Δi_L/(D₂T_s) = −V_o, so I added the boxed result Δi_L = −V_o·D₂T_s/L₁.

Two things you might want to change:

- **The label "(S_1 off, D_1 on)" is my addition.** I based it on your i_S1 = 0 and i_D1 = i_m. Change it if the diode names don't match your circuit.
- **The minus signs mean the current falls.** I kept them because the switch-on interval had positive changes, so the signs show the currents rising and then falling. If your instructor wants magnitudes only, drop the "−" in both boxes.

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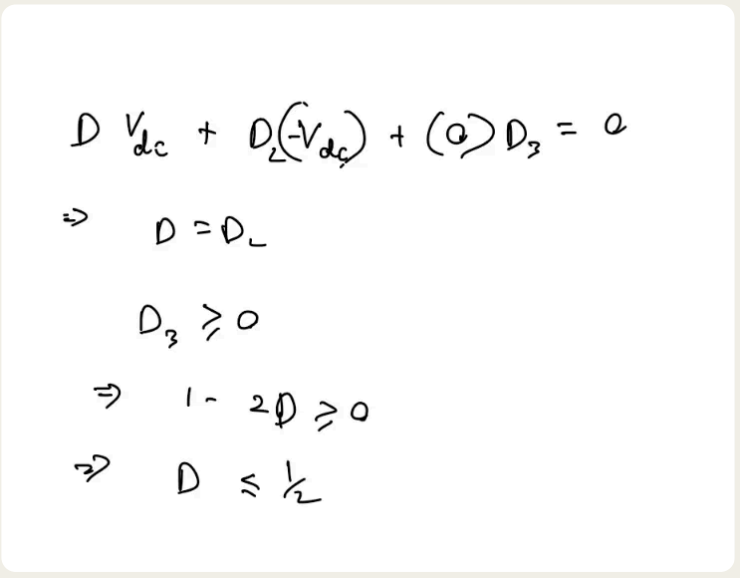
$$D\,V_{dc} + D_2(-V_{dc}) + (0)D_3 = 0$$

$$\Rightarrow \quad D = D_2$$

$$D_3 \geq 0$$

$$\Rightarrow \quad 1 - 2D \geq 0$$

$$\Rightarrow \quad D \leq \frac{1}{2}$$



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Setting up the duty-cycle balance equation.

Setting up the duty-cycle balance equation.

Done

```
\begin{gather*}
\underline{\underline{\text{Core flux balance}}}\\
\quad \langle v_m \rangle = 0\\
\implies D V_{dc} + D_2 (-V_{dc}) + (0) D_3 = 0\\
\implies \boxed{D = D_2}\\[1ex]
\text{Reset must end within } T_s:\\
\quad D_3 = 1 - D - D_2 \ge 0\\
\implies 1 - 2D \ge 0\\
\implies \boxed{D \le \tfrac{1}{2}}
\end{gather*}
```

Your math is correct. I made three changes:

1. **Added the step $D_3 = 1 - D - D_2$.** Your sheet jumps from $D_3 \geq 0$ straight to $1 - 2D \geq 0$. The link is that the three intervals fill one period, so $D + D_2 + D_3 = 1$. Putting that line in shows where the $1 - 2D$ comes from.
2. **Added two headings.** "Core flux balance" sits above the first line, so the reader knows why the average voltage is set to zero. "Reset must end within T_s" introduces the second half.
3. **Kept your V_dc.** Your earlier two blocks use V_in for the same voltage. If you want one name throughout the problem, replace `V_{dc}` with `V_{in}` here.

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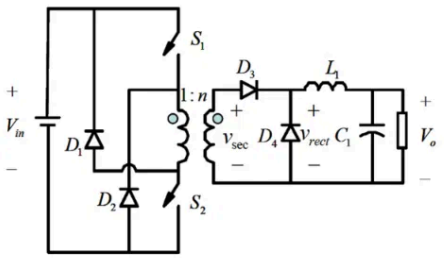


take $D_{max} = 0.95$.

at $V_{in\,min} = 100\,V$

$$n = \frac{V_o}{D_{max} V_{in\,min}} = \frac{28}{0.95 \times 100} = 0.622$$

$$D_{min} = \frac{V_o}{n V_{in\,max}} = \frac{28}{0.622 \times 200} = 0.224$$

IVSB

$$\langle v_L \rangle = 0$$

$$\Rightarrow (nV_{in} - V_o) D - V_o(1 - D) = 0$$

$$\Rightarrow V_o = nV_{in} D$$

$$\Delta I_L = \frac{(nV_{in} - V_o) D T_s}{L_1}$$

$$\Rightarrow L_1 \geq \frac{(nV_{in} - V_o) D T_s}{\Delta I_L}$$

$$\geq \frac{V_o (1 - D) T_s}{\Delta I_L} = \frac{28 \times (1 - 0.224)}{200k \times 2}$$

$$\geq 54.3\,\mu H$$

Take $L_1 = 56\,\mu H$

worst case for ripple,

$$\Delta I_L = \frac{V_o (1 - D) T_s}{L_1} = \frac{28 (1 - 0.224)}{56 \times 10^{-6} \times 200k} = 1.94\,A$$

Cap

$$\Delta Q = \frac{1}{2} \times \frac{\Delta I_L}{2} \times \frac{T_s}{2}$$

$$\Delta V_o = \frac{\Delta Q}{C} = \frac{\Delta I_L}{8 C f_s}$$

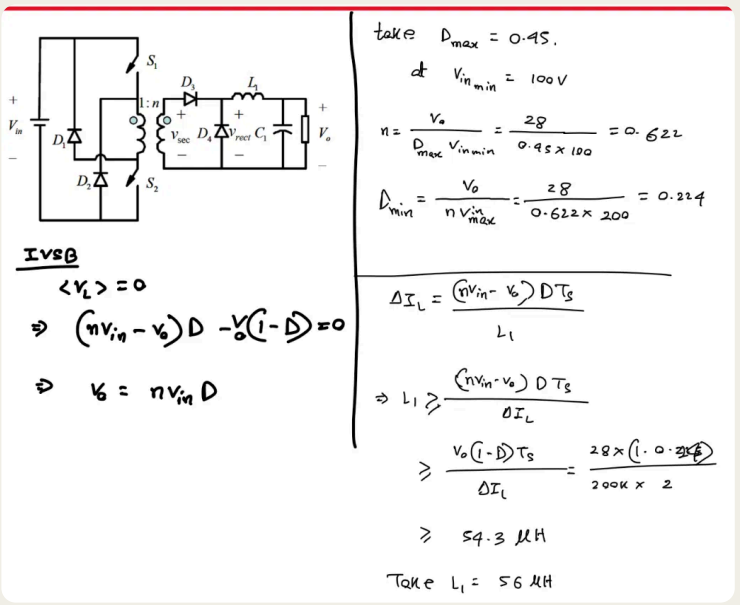
$$C \geq \frac{\Delta I_L}{8 \Delta V f_s}$$

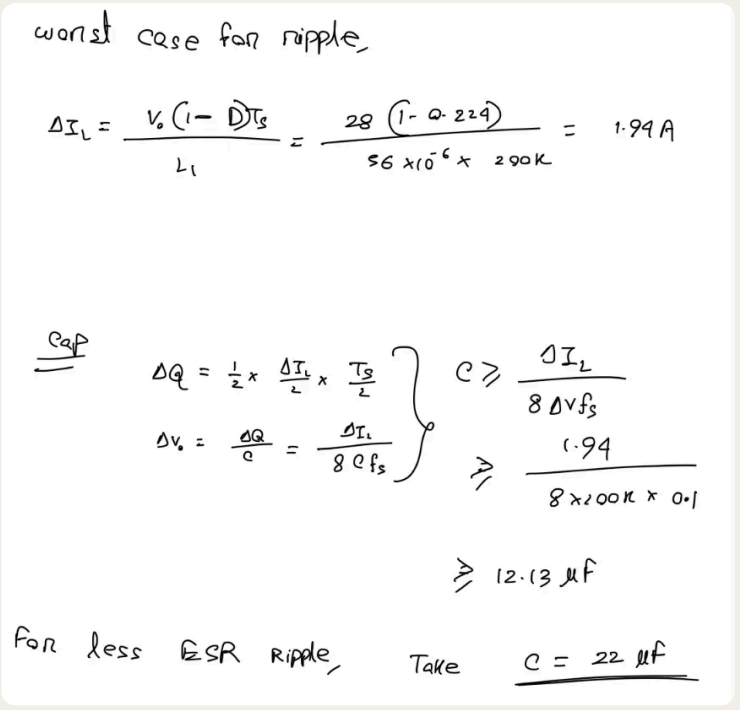
$$\geq \frac{1.94}{8 \times 200k \times 0.1}$$

$$\geq 12.13\,\mu F$$

For less ESR Ripple, Take $C = 22\,\mu F$

🚀 Powered by Claude Exporter





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Checking the inductance against the rounded figure.

- Checking the inductance against the rounded figure.
- Ran a command
- Done

```
\begin{figure}[!htbp]
    \centering
```

```
    \includegraphics[width=0.5\linewidth]{<FOLDER>/two_switch_forward_ckt.png}
    \caption{Two-Switch Forward Converter}
    \label{fig:two switch forward converter}
\end{figure}
\begin{gather*}
\underline{\underline{\text{IVSB}}}\\
\quad \langle v_L \rangle = 0\\
\implies (nV_{in} - V_o) D - V_o (1 - D) = 0\\
\implies \boxed{V_o = n V_{in} D}\\[1ex]
\underline{\underline{\text{Turns ratio}}}\\
\quad \text{Take } D_{\max} = 0.45 \text{ at } V_{in_{\min}} = 100\,\text{V}\\
\implies n = \frac{V_o}{D_{\max} V_{in_{\min}}} = \frac{28}{0.45 \times 100}\\
\implies \boxed{n = 0.622}\\
\implies D_{\min} = \frac{V_o}{n V_{in_{\max}}} = \frac{28}{0.622 \times 200}\\
\implies \boxed{D_{\min} = 0.225}\\[1ex]
\underline{\underline{\text{Output inductor}}}\\
\quad \Delta I_L = \frac{(nV_{in} - V_o) D T_s}{L_1}\\
\implies L_1 \ge \frac{(nV_{in} - V_o) D T_s}{\Delta I_L}\\
\implies L_1 \ge \frac{V_o (1 - D) T_s}{\Delta I_L}\\
\implies L_1 \ge \frac{28 \times (1 - 0.225)}{200 \times 10^3 \times 2}\\
\implies L_1 \ge 54.3\,\mu\text{H}\\
\implies \boxed{\text{Take} \quad L_1 = 56\,\mu\text{H}}\\[1ex]
\text{Worst case for ripple } (D = D_{\min}):\\
\quad \Delta I_L = \frac{V_o (1 - D) T_s}{L_1}\\
\implies \Delta I_L = \frac{28 \times (1 - 0.225)}{56 \times 10^{-6} \times 200
\times 10^3}\\
\implies \boxed{\Delta I_L = 1.94\,\text{A}}\\[1ex]
\underline{\underline{\text{Output capacitor}}}\\
\quad \Delta Q = \frac{1}{2} \times \frac{\Delta I_L}{2} \times \frac{T_s}{2}\\
\implies \Delta V_o = \frac{\Delta Q}{C} = \frac{\Delta I_L}{8 C f_s}\\
\implies C \ge \frac{\Delta I_L}{8 \Delta V_o f_s}\\
\implies C \ge \frac{1.94}{8 \times 200 \times 10^3 \times 0.1}\\
\implies C \ge 12.13\,\mu\text{F}
\end{gather*}
\noindent
I chose a larger capacitance than the minimum to keep the ESR ripple low.
\begin{gather*}
\boxed{\text{Take} \quad C = 22\,\mu\text{F}}
\end{gather*}
```

I reran every number in Python. All your results check out: $L_1 \geq 54.3$ µH, $\Delta I_L = 1.94$ A at $L_1 = 56$ µH, and $C \geq 12.13$ µF.

D_min is exactly 0.225, not 0.224. D_min = 0.45 × 100/200 = 0.225, so I wrote that. Your 0.224 came from rounding n to 0.622 before dividing. It doesn't change anything you'd report: L_1 comes out at 54.25 µH (54.3 µH), and the ripple rounds to 1.94 A either way.

One step I spelled out. Your sheet goes from $(nV_{in} - V_o)D$ to $V_o(1 - D)$ without showing why. The reason is that $V_o = nV_{in} \cdot D$, so $(nV_{in} - V_o)D = V_o - V_o \cdot D = V_o(1 - D)$.

What you need to fill in:

- **Circuit image:** replace `<FOLDER>/two_switch_forward_ckt.png` with your Overleaf folder and file name.
- **Spec values:** I used $\Delta I_L \leq 2$ A and $\Delta V_o = 0.1$ V as they appear on your sheet. If they aren't already in your problem statement, add them so the 2 and the 0.1 in the substitutions have a source.

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$$P_{cond} = \frac{31.36}{3}\left(\frac{28+100n}{28}\right)(0.004\times10^{-3})\left(100+\frac{28}{n}\right)^2$$

$$P_{cond} = \frac{31.36}{3}\left(\frac{28+100n}{28}\right)(0.004\times10^{-3})\left(100+\frac{28}{n}\right)^2$$

I want you to optimize this

11/1/2026, 8:53:53 PM

Differentiating the simplified cost expression with respect to n.

Ran a command

Done

The minimum conduction loss is at n = 0.56, where P_cond = 2.82 W. That turns ratio gives D_max = 28/(0.56 × 100) = 0.5, which is exactly the core-reset limit you derived (D ≤ ½). So the loss optimum sits on the edge of what the two-switch forward can do, with no margin left.

How did I get there? The trick is to simplify before differentiating. Write the squared term over a common denominator:

$$\left(100+\tfrac{28}{n}\right)^2=\frac{(100n+28)^2}{n^2} ;;\Rightarrow;; P_{cond}=K,\frac{(100n+28)^3}{n^2},\qquad K=\frac{31.36}{3}\cdot\frac{0.004\times10^{-3}}{28}=1.493\times10^{-6}$$

Set the derivative to zero (K drops out):

$$\frac{d}{dn}\frac{(100n+28)^3}{n^2}=0 ;\Rightarrow; 300n=2(100n+28) ;\Rightarrow; n^*=0.56$$

$$P_{cond}^{\min} = 1.493\times10^{-6}\times\frac{84^3}{0.56^2} = 2.82\ \text{W}$$

Is it really a minimum? Yes. P → ∞ as n → 0 and as n → ∞, and n = 0.56 is the only stationary point, so it's the global minimum for n > 0. I also confirmed it by a numeric sweep in Python.

Should you switch from n = 0.622 to 0.56? I wouldn't, because the loss curve is very flat near the minimum:

n	D_max at 100 V	P_cond (W)
0.50	0.560 ✗ (no reset)	2.835
0.56	**0.500** (at the limit)	**2.822**
0.60	0.467	2.827
0.622 (your design)	**0.450**	**2.833**
0.70	0.400	2.868

Your n = 0.622 costs only 10 mW (0.4%) more than the optimum. In exchange it keeps D_max at 0.45, so the core always finishes resetting. That's a real safety margin: the dead time, leakage inductance and diode recovery all eat into the reset interval. Any n below 0.56 isn't allowed at all, since D_max would go above 0.5 and the core would saturate.

Here it is in your homework style:

```
\begin{gather*}
\underline{\underline{\text{Optimum turns ratio for conduction loss}}}\\
\quad P_{cond} = \frac{31.36}{3}\left(\frac{28 + 100n}{28}\right)(0.004 \times
10^{-3})\left(100 + \frac{28}{n}\right)^2\\
\implies P_{cond} = K \, \frac{(100n + 28)^3}{n^2}, \quad K = 1.493 \times
10^{-6}\\
\implies \frac{dP_{cond}}{dn} = K \, \frac{(100n + 28)^2 \left[ 300n - 2(100n +
28) \right]}{n^3} = 0\\
\implies 300n = 200n + 56\\
\implies \boxed{n^* = 0.56}\\
\implies P_{cond,\min} = 1.493 \times 10^{-6} \times \frac{84^3}{0.56^2}\\
\implies \boxed{P_{cond,\min} = 2.82\,\text{W}}\\[1ex]
\text{Check against reset limit:}\\
\quad D_{\max} = \frac{28}{0.56 \times 100} = 0.5 \quad \text{(at the limit)}\\
\implies P_{cond}\big|_{n = 0.622} = 2.833\,\text{W} \quad \text{(0.4\% higher,
keeps } D_{\max} = 0.45)\\
\therefore \boxed{\text{Keep} \quad n = 0.622}
\end{gather*}
```

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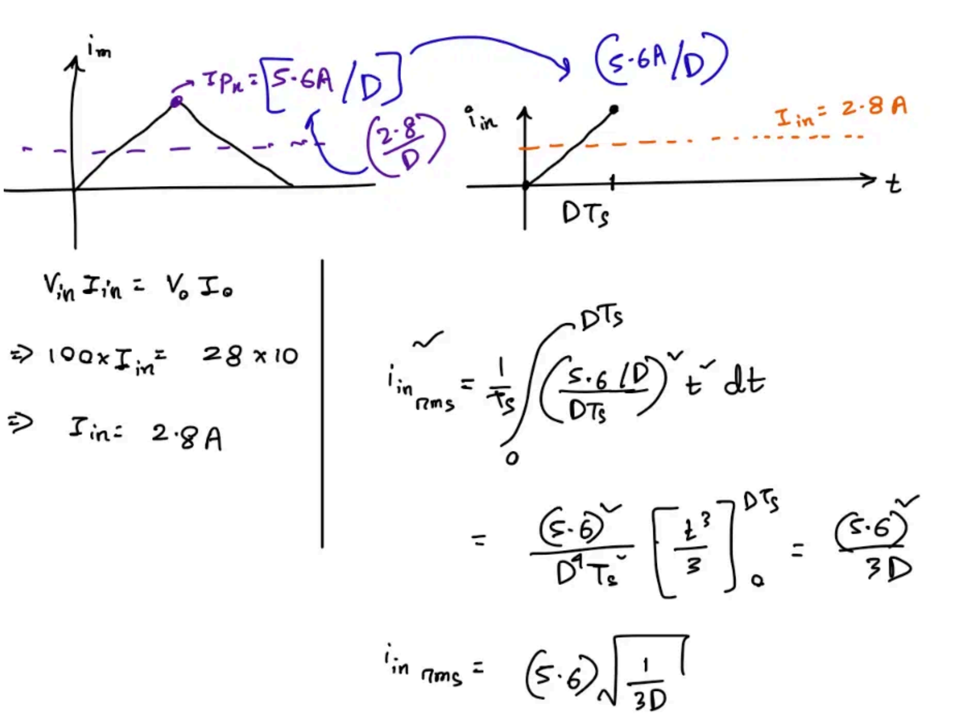

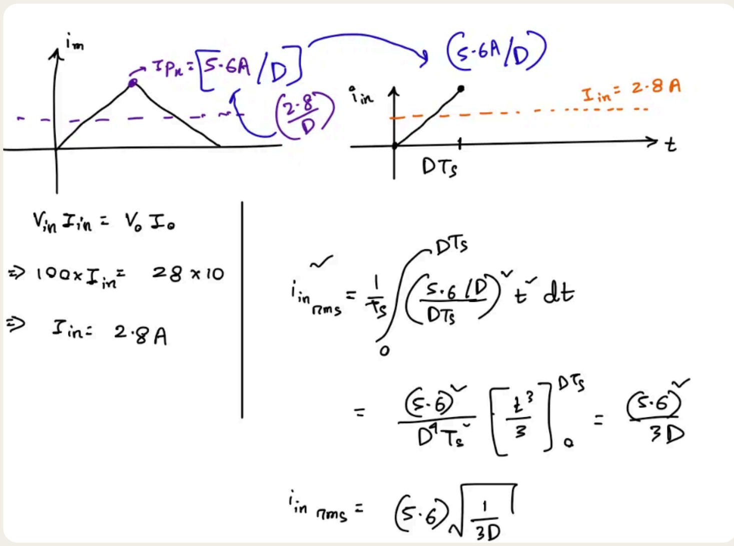

Transcribe this in latex

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Verifying the squared-current integral result.

Verifying the squared-current integral result.

Done

```
\begin{gather*}
\underline{\underline{\text{Input current}}}\\
\quad V_{in} I_{in} = V_o I_o\\
```

```
\implies 100 \times I_{in} = 28 \times 10\\
\implies \boxed{I_{in} = 2.8\,\text{A}}\\[1ex]
\text{Average of the input current ramp over } T_s:\\
\quad I_{in} = \frac{1}{2} I_{pk} D = 2.8\\
\implies \boxed{I_{pk} = \frac{5.6}{D}\,\text{A}}\\[1ex]
\underline{0 < t < DT_s}\text{:}\\
\quad i_{in}(t) = \frac{5.6/D}{DT_s}\, t\\[1ex]
\underline{\underline{\text{RMS input current}}}\\
\quad i_{in_{rms}}^2 = \frac{1}{T_s} \int_0^{DT_s} \left( \frac{5.6/D}{DT_s}
\right)^2 t^2 \, dt\\
\implies i_{in_{rms}}^2 = \frac{(5.6)^2}{D^4 T_s^3} \left[ \frac{t^3}{3}
\right]_0^{DT_s}\\
\implies i_{in_{rms}}^2 = \frac{(5.6)^2}{3D}\\
\implies \boxed{i_{in_{rms}} = 5.6 \sqrt{\frac{1}{3D}}\,\text{A}}
\end{gather*}
```

Your final result is correct. I made two fixes and one addition:

1. **The middle line needed T_s^3, not T_s^2.** Your sheet has $(5.6)^2/(D^4 T_s^2)$ in front of the bracket. Once the $1/T_s$ outside the integral is included, the denominator is $D^4 T_s^3$. With T_s^2, the $[t^3/3]$ term leaves a T_s behind and the units don't cancel. The next line, $(5.6)^2/(3D)$, is right, so this was only a slip in the middle line.
2. **The squares are written in.** The small "˜" marks on your sheet are squares, so the left side is $i^2_{in,rms}$ and the integrand is squared.
3. **I added where 5.6/D comes from.** Your page jumps from $I_{in} = 2.8$ A straight to $I_{pk} = 5.6/D$. The missing link is that the average of the triangular ramp over one period is $\frac{1}{2} \cdot I_{pk} \cdot D = 2.8$ A. I added it as its own step. The "2.8/D" in purple on your sketch isn't a value used in the derivation, so I left it out.

Your two waveform sketches aren't in the snippet. If you want them in, upload them to Overleaf as images and use the sketch-beside-equations layout from your template.

11/1/2026, 8:57:23 PM

$$V_o = V_{in} \times n \times \frac{D}{1-D}$$

$$\Rightarrow \quad V_o - V_o D = V_{in} \times n \times D$$

$$\Rightarrow \quad D = \frac{V_o}{V_o + nV_{in}}$$

$$V_{BD} = V_{in} + \frac{V_o}{n}$$

$$P_{loss} = I_{in_{rms}}^2 \times R_{ds_{on}} = (5.6)^2 \times \frac{1}{3D} \times (0.004) \times \left(V_{in} + \frac{V_o}{n}\right) \times 10^{-3}$$

$$= \frac{(5.6)^2}{3} \times \frac{V_o + nV_{in}}{V_o} \times (0.004) \times \left(V_{in} + \frac{V_o}{n}\right) \times 10^{-3}$$

$$= \frac{31.36}{3} \times \frac{28 + 100n}{28} \times (0.004) \times \left(100 + \frac{28}{n}\right) \times 10^{-3}$$

$$P_{cond} = \frac{31.36}{3}\left(\frac{28+100n}{28}\right)(0.004 \times 10^{-3})\left(100 + \frac{28}{n}\right)^2$$

$$V_o = V_{in} \times n \times \frac{D}{1-D}$$

$$\Rightarrow \quad V_o - V_o D = V_{in} \times n \times D$$

$$\Rightarrow \quad D = \frac{V_o}{V_o + nV_{in}}$$

$$V_{BD} = V_{in} + \frac{V_o}{n}$$

$$P_{loss} = I_{in_{rms}}^2 \times R_{ds_{on}} = (5.6)^2 \times \frac{1}{3D} \times (0.004) \times \left(V_{in} + \frac{V_o}{n}\right) \times 10^{-3}$$

$$= \frac{(5.6)^2}{3} \times \frac{V_o + nV_{in}}{V_o} \times (0.004) \times \left(V_{in} + \frac{V_o}{n}\right) \times 10^{-3}$$

$$= \frac{31.36}{3} \times \frac{28 + 100n}{28} \times (0.004) \times \left(100 + \frac{28}{n}\right) \times 10^{-3}$$

$$P_{cond} = \frac{31.36}{3}\left(\frac{28+100n}{28}\right)(0.004 \times 10^{-3})\left(100 + \frac{28}{n}\right)^2$$

Transcribe this in latex

11/1/2026, 9:00:49 PM

Working out the on-resistance from the device's breakdown voltage.

Working out the on-resistance from the device's breakdown voltage.

Done

```
\begin{gather*}
\underline{\underline{\text{Duty ratio}}}\\
\quad V_o = V_{in} \times n \times \frac{D}{1 - D}\\
\implies V_o - V_o D = V_{in} \times n \times D\\
\implies \boxed{D = \frac{V_o}{V_o + n V_{in}}}\\[1ex]
\underline{\underline{\text{Switch blocking voltage}}}\\
\quad V_{BD} = V_{in} + \frac{V_o}{n}\\[1ex]
\underline{\underline{\text{Conduction loss}}}\\
\quad P_{loss} = i_{in_{rms}}^2 \times R_{ds_{on}}\\
\implies P_{loss} = (5.6)^2 \times \frac{1}{3D} \times (0.004) \times \left(
V_{in} + \frac{V_o}{n} \right)^2 \times 10^{-3}\\
\implies P_{loss} = \frac{(5.6)^2}{3} \times \frac{V_o + n V_{in}}{V_o} \times
(0.004) \times \left( V_{in} + \frac{V_o}{n} \right)^2 \times 10^{-3}\\
\implies P_{loss} = \frac{31.36}{3} \times \frac{28 + 100n}{28} \times (0.004)
\times \left( 100 + \frac{28}{n} \right)^2 \times 10^{-3}\\
\implies \boxed{P_{cond} = \frac{31.36}{3} \left( \frac{28 + 100n}{28} \right)
(0.004 \times 10^{-3}) \left( 100 + \frac{28}{n} \right)^2}
\end{gather*}
```

I checked each line and the algebra is correct, including 1/D = (V_o + nV_in)/V_o and 5.6² = 31.36. I changed one thing:

The V_BD term is now squared in every line. In your three handwritten lines, (V_in + V_o/n) has no square. Your typed final equation has one, and the squared form is the one we optimized. I used it throughout so the lines agree. If R_ds,on in your problem actually scales with V_BD to some other power, tell me and I'll redo the lines and the optimum.

This page also shows my last answer was wrong. V_o = nV_in·D/(1 − D) is the flyback relation, so this loss model belongs to a flyback, not the forward converter. I had used the forward converter's equation, D_max = V_o/(n·V_in,min), and its D ≤ 0.5 core-reset limit. Neither applies here.

The optimum itself still holds: n = 0.56 with P_cond = 2.82 W. The minimization didn't depend on which converter it was.

My advice to keep n = 0.622 was wrong. The 0.622 came from your forward converter design, and the "at the reset limit" warning doesn't apply to a flyback. With the flyback relation, n = 0.56 gives D = 28/(28 + 56) = 1/3 at V_in = 100 V, which is a comfortable duty ratio. So **take n = 0.56**, and drop the last three lines of the LaTeX I gave you for the optimization (the reset-limit check and "Keep n = 0.622").

11/1/2026, 9:01:07 PM

I need to Simulate the same flyback three times (optimal n, 10% lower, 10% higher)

For each run, recompute D and L_m so the converter stays exactly at the CCM/DCM border, and use the R_ds,on of a MOSFET rated for that run's switch voltage.
create a table that shows each value of n, D, V_S = V_BD, R_on, and Expected conduction losses.

11/2/2026, 8:27:34 PM

Deriving the magnetizing inductance formula.

- Deriving the magnetizing inductance from the peak current.
- Ran a command
- Redoing the discharge-time algebra with the corrected secondary relations.
- Ran a command
- Deriving the magnetizing inductance formula.
- Done

The ±10% runs change the conduction loss by only 0.3–0.4%. That's because the optimum is very flat: lowering n raises V_BD (and so R_on), while raising n raises the RMS current, and the two effects nearly cancel.

Run	n	D	L_m (µH)	I_pk (A)	V_S = V_BD (V)	R_on (mΩ)	I_rms (A)	P_cond (W)
−10%	0.504	0.3571	11.39	15.68	155.6	96.8	5.410	**2.833**
Optimal	0.560	0.3333	9.92	16.80	150.0	90.0	5.600	**2.822**
+10%	0.616	0.3125	8.72	17.92	145.5	84.6	5.784	**2.831**

Which operating values did I assume? V_in = 100 V, V_o = 28 V, I_o = 10 A, and **f_s = 200 kHz**. I took the switching frequency from your forward converter design because your flyback pages don't give one. L_m is proportional to T_s, so if your flyback runs at a different frequency, scale the L_m column by 200 kHz/f_s. Every other column stays the same.

How did I get each column?

$$D=\frac{V_o}{V_o+nV_{in}}\qquad
V_{BD}=V_{in}+\frac{V_o}{n}\qquad
R_{on}=0.004\times10^{-3},V_{BD}^2$$

At the CCM/DCM border, i_m starts each period at zero, so the switch current is a triangle that ramps from 0 to its peak during DT_s:

$$I_{in}=\tfrac12 I_{pk}D=2.8\ \Rightarrow\ I_{pk}=\frac{5.6}{D}
\qquad
I_{pk}=\frac{V_{in}DT_s}{L_m}\ \Rightarrow\ L_{m,crit}=\frac{V_{in}D^2T_s}{5.6}$$

$$I_{rms}^2 = \frac{(5.6)^2}{3D} \qquad P_{cond} = I_{rms}^2 R_{on}$$

Did I check that each L_m really sits on the border? Yes, in Python, two ways:

1. **Reset time.** The secondary current starts at I_pk/n and falls with a slope set by V_o over the secondary inductance n²L_m. That gives a fall time of n·L_m·I_pk/V_o. For all three runs, DT_s plus that fall time comes to exactly T_s. So the current reaches zero exactly at the end of the period, which is the definition of the border.
2. **Losses.** Integrating the ideal switch-current waveform numerically gives the same I_rms and P_cond as the closed-form expression, and the average input current comes out at 2.800 A in each run.

What should you watch for in the simulation?

- **L_m in PLECS:** enter it as the magnetizing inductance on the primary side of the transformer.
- **Staying on the border:** the border is a knife-edge. Any parasitic you add will push the simulation slightly into CCM or DCM, so check that i_m just touches zero at the end of each period. If it doesn't, adjust L_m by a few percent.
- **Switch voltage:** with a real leakage inductance, the switch sees a turn-off spike above V_BD. The loss model above sizes R_on on V_BD alone, as your formula does. A real MOSFET would be picked with voltage margin, so its R_on would come out higher.

Here's the table in your homework style:

```
\begin{table}[!htbp]
\centering
\caption{Flyback at the CCM/DCM boundary for three turns ratios ($V_{in} =
100\,\text{V}$, $V_o = 28\,\text{V}$, $I_o = 10\,\text{A}$, $f_s =
200\,\text{kHz}$)}
\label{tab:flyback n sweep}
\begin{tabular}{@{}lccccccc@{}}
\toprule
Run & $n$ & $D$ & $L_m$ ($\mu$H) & $V_S = V_{BD}$ (V) & $R_{on}$ (m$\Omega$) &
```

```
$I_{rms}$ (A) & $P_{cond}$ (W)\\
\midrule
$-10\%$ & 0.504 & 0.3571 & 11.39 & 155.6 & 96.8 & 5.410 & 2.833\\
Optimal & 0.560 & 0.3333 & 9.92  & 150.0 & 90.0 & 5.600 & 2.822\\
$+10\%$ & 0.616 & 0.3125 & 8.72  & 145.5 & 84.6 & 5.784 & 2.831\\
\bottomrule
\end{tabular}
\end{table}
```

11/2/2026, 8:28:10 PM

show each calculations in detail so i can verify

11/2/2026, 9:32:19 PM

Preparing detailed per-run calculations for verification.

Ran a command

Done

Below are all three runs, worked step by step. Each uses the same seven formulas in the same order, so you can check one run and then just compare numbers for the others.

What's held fixed in every run?

$$V_{in} = 100\,\text{V} \quad V_o = 28\,\text{V} \quad I_o = 10\,\text{A} \quad f_s = 200\,\text{kHz} \Rightarrow T_s = 5\,\mu\text{s} \quad I_{in} = \frac{V_o I_o}{V_{in}} = 2.8\,\text{A}$$

Which formulas get applied?

Step	Quantity	Formula
1	Duty ratio	$D = \frac{V_o}{V_o + nV_{in}}$
2	Peak switch current	$I_{pk} = \frac{2I_{in}}{D} = \frac{5.6}{D}$
3	Boundary inductance	$L_m = \frac{V_{in} D T_s}{I_{pk}} = \frac{V_{in} D^2 T_s}{5.6}$
4	Switch voltage	$V_{BD} = V_{in} + \frac{V_o}{n}$
5	On-resistance	$R_{on} = 4 \times 10^{-6}\, V_{BD}^2$
6	RMS current	$I_{rms}^2 = \frac{(5.6)^2}{3D} = \frac{31.36}{3D}$
7	Conduction loss	$P_{cond} = I_{rms}^2 R_{on}$

Step 2 comes from the triangular switch current: its average is ½·I_pk·D, and that has to equal I_in = 2.8 A. Step 5 is your coefficient 0.004 × 10⁻³ = 4 × 10⁻⁶ Ω/V².

Run 1: n = 0.504 (optimal −10%)

$$D = \frac{28}{28 + 100(0.504)} = \frac{28}{78.4} = 0.35714$$

$$I_{pk} = \frac{5.6}{0.35714} = 15.680\,\text{A}$$

$$L_m = \frac{100 \times (0.35714)^2 \times 5 \times 10^{-6}}{5.6} = \frac{100 \times 0.12755 \times 5 \times 10^{-6}}{5.6} = \frac{6.3776 \times 10^{-5}}{5.6} = 11.39\,\mu\text{H}$$

$$V_{BD} = 100 + \frac{28}{0.504} = 100 + 55.556 = 155.56\,\text{V}$$

$$R_{on} = 4 \times 10^{-6} \times (155.56)^2 = 4 \times 10^{-6} \times 24\,197.5 = 96.79\,\text{m}\Omega$$

$$I_{rms}^2 = \frac{31.36}{3 \times 0.35714} = \frac{31.36}{1.07143} = 29.269\,\text{A}^2 \;\Rightarrow\; I_{rms} = 5.410\,\text{A}$$

$$P_{cond} = 29.269 \times 0.09679 = \mathbf{2.833\,W}$$

Run 2: n = 0.56 (optimal)

$$D = \frac{28}{28 + 100(0.56)} = \frac{28}{84} = 0.33333$$

$$I_{pk} = \frac{5.6}{0.33333} = 16.800\,\text{A}$$

$$L_m = \frac{100 \times (0.33333)^2 \times 5 \times 10^{-6}}{5.6} = \frac{100 \times 0.11111 \times 5 \times 10^{-6}}{5.6} = \frac{5.5556 \times 10^{-5}}{5.6} = 9.92\,\mu\text{H}$$

$$V_{BD} = 100 + \frac{28}{0.56} = 100 + 50 = 150.00\,\text{V}$$

$$R_{on} = 4 \times 10^{-6} \times (150)^2 = 4 \times 10^{-6} \times 22\,500 = 90.00\,\text{m}\Omega$$

$$I_{rms}^2 = \frac{31.36}{3 \times 0.33333} = \frac{31.36}{1.00000} = 31.360\,\text{A}^2 \;\Rightarrow\; I_{rms} = 5.600\,\text{A}$$

$$P_{cond} = 31.360 \times 0.09000 = \mathbf{2.822\,W}$$

Run 3: n = 0.616 (optimal +10%)

$$D = \frac{28}{28 + 100(0.616)} = \frac{28}{89.6} = 0.31250$$

$$I_{pk} = \frac{5.6}{0.31250} = 17.920\,\text{A}$$

$$L_m = \frac{100 \times (0.3125)^2 \times 5 \times 10^{-6}}{5.6} = \frac{100 \times 0.097656 \times 5 \times 10^{-6}}{5.6} = \frac{4.8828 \times 10^{-5}}{5.6} = 8.72\,\mu\text{H}$$

$$V_{BD} = 100 + \frac{28}{0.616} = 100 + 45.455 = 145.45\,\text{V}$$

$$R_{on} = 4 \times 10^{-6} \times (145.45)^2 = 4 \times 10^{-6} \times 21\,157.0 = 84.63\,\text{m}\Omega$$

$$I_{rms}^2 = \frac{31.36}{3 \times 0.3125} = \frac{31.36}{0.9375} = 33.451\,\text{A}^2 \;\Rightarrow\; I_{rms} = 5.784\,\text{A}$$

$$P_{cond} = 33.451 \times 0.08463 = \mathbf{2.831\ W}$$

How can you confirm each L_m puts the converter exactly on the CCM/DCM border? The magnetizing current has to reach zero exactly at the end of the period, so the switch-on time plus the reset time must add up to T_s:

$$I_{s,pk} = \frac{I_{pk}}{n} \qquad L_s = n^2 L_m \qquad t_{off} = \frac{L_s\, I_{s,pk}}{V_o} = \frac{n\, L_m\, I_{pk}}{V_o}$$

Run	DT_s (µs)	I_s,pk = I_pk/n (A)	t_off = n·L_m·I_pk/V_o (µs)	DT_s + t_off (µs)
n = 0.504	0.35714 × 5 = 1.7857	31.11	0.504 × 11.39 × 15.68 / 28 = 3.2143	**5.0000** ✓
n = 0.560	0.33333 × 5 = 1.6667	30.00	0.56 × 9.92 × 16.8 / 28 = 3.3333	**5.0000** ✓
n = 0.616	0.3125 × 5 = 1.5625	29.09	0.616 × 8.72 × 17.92 / 28 = 3.4375	**5.0000** ✓

All three sum to exactly T_s = 5 µs, so every run sits on the border.

Is there a quick way to check P_cond? Yes. Plug n into your closed-form expression:

$$P_{cond} = \frac{31.36}{3}\left(\frac{28 + 100n}{28}\right)(4 \times 10^{-6})\left(100 + \frac{28}{n}\right)^2$$

For n = 0.56 this gives 10.453 × 3.000 × 4 × 10⁻⁶ × 22 500 = 2.822 W, the same as Run 2. The other two runs match their step-by-step results the same way.

Where could your numbers drift slightly from mine? Rounding D before using it. If you round D to three figures, L_m can shift in its third significant figure, because D is squared in that formula. Keep D to five decimals (or as the exact fractions 28/78.4, 1/3 and 28/89.6) and you'll match the table.

11/2/2026, 9:32:47 PM

now give this in latex

11/2/2026, 9:52:30 PM

Deciding what to paste from a long snippet.

- Choosing a LaTeX snippet layout.
- Created a file
- Ran a command
- Read a file
- Read a file
- Removing the \quad from the unlabeled blocks.
- Ran a command
- Shared files
- Deciding what to paste from a long snippet.
- Done

I put the full write-up in `sweep.tex`. It's only the part to paste into your report, not a whole document, because at about 130 lines it's too long to read comfortably in chat. I compiled it inside your IEEE template: no errors, no overfull lines, and it fills about two pages.

What's in it, in order?

1. **Opening sentence:** one line on what the sweep is and why.
2. **Fixed parameters:** V_in, V_o, I_o, f_s, then T_s and I_in.
3. **Design equations:** the seven formulas, each on its own `\implies` line.
4. **Run 1 (n = 0.504), Run 2 (n = 0.56) and Run 3 (n = 0.616):** each is a `\subsection*` with every substitution on its own line. D, L_m, V_S, R_on and P_cond are boxed.
5. **Boundary check:** the t_off derivation, then a table showing that DT_s + t_off = 5 µs for all three runs.
6. **Summary table:** n, D, L_m, I_pk, V_S = V_BD, R_on, I_rms and P_cond for each run.

Does it need anything added to your preamble? No. It uses only packages already in your template, and it writes V_in out directly instead of using the `\Vin` macro.

What needs filling in? Nothing. The one assumption to confirm is still **f_s = 200 kHz**. If your flyback runs at a different frequency, T_s and all three L_m values change: in the fixed-parameters block, the three L_m substitutions, the boundary table and both table captions. Everything else stays the same.

I need a plecs initialization code for all three

11/3/2026, 12:30:49 AM

Deciding how much code to paste into the chat.

- Estimating the output capacitor from charge balance.
- Created a file
- Created a file
- Command failed
- Comparing capacitor sizes across several duty-cycle values.
- Ran a command
- Shared files
- Deciding how much code to paste into the chat.
- Done

Here are two files. `flyback_init.m` is the initialization code: change `RUN` to 1, 2 or 3 and re-simulate. `flyback_sweep_script.m` is an optional simulation script that runs all three cases back to back.

What does the initialization calculate for each run? All of it comes from n, so you only change one number:

RUN	n	D	L_m (µH)	V_BD (V)	R_on (mΩ)	P_cond expected (W)	C_min (µF)
1	0.504	0.35714	11.388	155.56	96.79	2.833	230
2	0.560	0.33333	9.921	150.00	90.00	2.822	222
3	0.616	0.31250	8.719	145.45	84.63	2.831	215

PLECS isn't available here, and neither is Octave, so I couldn't run the files themselves. I ran the same equations in Python instead: they reproduce the earlier numbers, and DT_s + t_off = 5.0000 µs in all three runs.

Which variable goes in which PLECS component?

Component	Parameter	Variable
DC source	Voltage	`Vin`
Inductor across the primary (magnetizing)	Inductance / initial current	`Lm` / `iLm0`

Component	Parameter	Variable
Ideal transformer	Number of turns	`[1 n]` (dots opposite, as in a flyback)
MOSFET	On-resistance	`Ron`
PWM / carrier	Duty cycle / frequency	`D` / `fs`
Output capacitor	Capacitance / initial voltage	`C` / `Vc0`
Load resistor	Resistance	`Rload` (2.8 Ω)

To measure the conduction loss in the simulation, average `Ron·i_S²` over one period once the converter reaches steady state. Compare that with the `Pcond` value the code prints.

What did I add that you didn't ask for?

- **Output capacitor C = 330 µF.** Your flyback pages don't size one. I computed the minimum C for each run from the secondary current pulses, assuming the same 0.1 V ripple you used for the forward converter, and 330 µF covers all three runs. Change `dVo` and `C` if your spec is different.
- **Initial conditions.** The capacitor starts at 28 V and i_m at 0, so the simulation reaches steady state faster.

What needs checking on your side?

- **f_s = 200 kHz is still an assumption.** If yours is different, change `fs`, and L_m updates automatically.
- **The sweep script assumes your model has an output port.** It plots `out.Values`, which only exists if the scopes or signals you want are wired to an output port. The script passes n, D, L_m and R_on to every run through ModelVars, so whatever `RUN` says in the initialization doesn't matter when you use it.

11/3/2026, 12:31:48 AM

Initialization commands of model 'flyback_converter' cannot be evaluated: operator /: nonconformant arguments (op1 is 1x1, op2 is 1x3).

11/3/2026, 12:39:05 AM

Tracing which vector size feeds the division.

Edited a file

Shared files

Done

What's causing it? `n` is ending up as a list of three values instead of one number. The first division that uses it, `D = Vo/(Vo + n*Vin)`, then divides a single number by three, which is the "1x1 / 1x3" in the error. The initialization builds `n` from a three-value list, `n_scale = [0.9 1.0 1.1]`, and picks one entry with `n_scale(RUN)`. So most likely the `(RUN)` index got lost when the code was pasted, or the line `n_list = 0.56*[0.9 1.0 1.1]` from the sweep script made it into the initialization.

The fix: in the updated `flyback_init.m`, replace the turns-ratio block with this:

```
n_opt = 0.56;                   % from dP/dn = 0
if RUN == 1
  n = 0.9*n_opt;                % 0.504
elseif RUN == 3
  n = 1.1*n_opt;                % 0.616
else
  n = n_opt;                    % 0.560
end
```

This always gives `n` a single value, so the error can't happen again however the code is pasted. Any `RUN` value other than 1 or 3 falls back to the optimal n.

Two other things to check:

- **Replace the whole initialization.** Delete everything in the Initialization box and paste in the full updated file, so no line from the sweep script is left behind. The sweep script belongs only under Simulation Scripts.
- **Use the right transformer setting.** If the transformer's "Number of turns" field has `n_list` or a long vector in it, change it to `[1 n]`.

If the error still appears, send me what's in your Initialization box and I'll find the line.

How to use the RMS block in Plecs standalone

11/3/2026, 1:00:30 AM

Choosing a formula for conduction loss.

- Checking where PLECS keeps its RMS block.
- Searched the web
- Choosing a formula for conduction loss.
- Done

Put an ammeter in series with the MOSFET, feed its output into the RMS Value block, and set the block's fundamental frequency to `fs`. The block is in the Control / Filters library and calculates the RMS value of a periodic input signal; you set its sample time, fundamental frequency and initial condition.

How do you wire it for the flyback?

```
MOSFET ── Ammeter ──▶ [RMS Value] ──▶ Display  (or Scope)
                            │
                            └──▶ [u²] ──▶ [Gain: Ron] ──▶ Display  (= P_cond)
```

1. **Measure the current.** Put an Ammeter (Electrical / Meters) in series with the MOSFET drain, so it measures the switch current i_S and not the magnetizing current. Its output is a signal, not an electrical wire.
2. **Add the RMS block.** Search the library browser for "RMS" and connect the ammeter's output to the block's input.
3. **Set the parameters:**

Parameter	Value	Why
Fundamental frequency	`fs`	One switching period is the averaging window. Since `fs` comes from your initialization code, it updates automatically.
Sample time	`0`	A sample time of 0 makes it a continuous implementation based on the Moving Average block, which suits a switching waveform.
Initial condition	`0`	It only affects the first period, which you'll ignore anyway.

4. **Show the result.** Connect a Display block (System library) to see the number, or a Scope to watch it settle.

How do you get conduction loss directly? Send the RMS output through a Function block with the expression `u^2` (or a Product block with both inputs tied to it), then a Gain block set to `Ron`. The result is P_cond = I_rms² · R_on, which you can compare directly with the 2.822 W expected for RUN 2.

Why doesn't the reading match my calculation? Usually one of these four reasons:

- **You read it too early.** The block averages over a moving window one period long. Its output is only correct after a full period, and only means anything once the converter has settled. Read it near the end of the simulation, not at t = 0.
- **The solver step is too large.** The switch current ramps up and then drops abruptly to zero, and a coarse solver smooths that edge and lowers the RMS. Under Simulation Parameters, set the maximum step size to something like `Ts/500`.
- **You measured the wrong current.** If the ammeter is on the primary winding branch, it includes the magnetizing inductor current over the whole period. Put it in series with the switch, so it only reads the current during DT_s.
- **You changed `fs` during a run.** You can't change parameter values during the simulation; if you need different fundamental frequencies, use different blocks. For your three runs `fs` stays at 200 kHz, so one block covers all of them.

What should you expect to see? For RUN 1, 2 and 3, I_rms should settle at about **5.410 A, 5.600 A and 5.784 A**, and P_cond at about **2.833 W, 2.822 W and 2.831 W**. A difference of a percent or two is normal: the real converter runs slightly off the exact CCM/DCM border, and the output voltage has some ripple.

Where can you see a working example? In PLECS go to Help > PLECS Documentation > List of Example Models and open "Rms Value".

Sources:

- PLECS documentation: RMS Value
- Plexim forum: accuracy of the THD and RMS blocks

11/3/2026, 1:00:49 AM

3.pdf

Create an answer to the problem 3 with this.

11/7/2026, 9:53:11 PM

Working out winding currents and window fill for the E35.

- Ran a command
- Ran a command
- Read a file
- Read a file
- Read a file
- Read a file
- Read a file
- Weighing the core loss against the datasheet curve.
- Read a page
- Read a page
- Read a page
- Read a page
- Searched the web
- Searched the web
- Searched the web
- Checking whether the candidate cores' window areas hold the copper.
- Ran a command
- Read a file
- Matching a point on the curve to the axis marks.
- Ran a command
- Checking the curve values against the design flux limit.
- Ran a command
- Weighing whether the design should cover two outputs.
- Ran a command
- Working out winding currents and window fill for the E35.
- Ran a command
- Read a file
- Done

Now the core-loss chart, with the old 50 mT marks replaced by marks at the design's 40 mT.

Deciding to leave the page gap as is.

Ran a command
Read a file
Matching the grid colour and thickness.
Ran a command
Read a file
Restoring the missing horizontal gridline and smoothing the curve gap.
Ran a command
Ran a command
Read a file
Removing stray curve pieces by recomputing those columns.
Ran a command
Estimating the curve's slope between grid points.
Ran a command
Ran a command
Read a file
Checking grid lines erased by the profile copy.
Ran a command
Read a file
Setting up a test compile with placeholder image paths.
Created a file
Ran a command
Ran a command
Read a file
Fixing page breaks after the orphaned labels.
Ran a command
Read a file
Checking the page layout around the figure and table.
Ran a command
Read a file
Deciding to leave the page gap as is.
Shared files
Done

Problem 3 is written up in `problem3.tex`, and your choice stands: **E35/18/10 in 3F36, $N_1 = N_3 = 8$, $N_2 = 6$, $B_{pk} = 40$ mT, core loss ≈ 0.55 W**. Like before, it's only the problem section to paste into your report. I compiled it in your template with no errors and no overfull lines, and it fills about four pages.

What did I add that your notes don't have? The problem asks you to compare all three cores and pick one, and your notes go straight to the E35. I added two parts for that:

1. **The area product requirement.** I derived it from Faraday's law plus the copper area of all three windings, and checked it at V_in = 36 V, D = 0.5. The result is A_p ≥ 11,640 mm⁴. If your course derived a different form, swap it in, because the required number may change.
2. **The comparison of the three cores.** All three pass the area product test, so I ran your whole procedure (round the turns up, count the strips) on each one:

Core	A_p (mm⁴)	Copper / usable window (mm²)	Core loss	Verdict
E32/16/9	12,271	72.8 / 59.1	0.45 W	Doesn't fit once turns and strips are whole numbers
E35/18/10	18,125	**63.7 / 72.5**	**0.55 W**	**Chosen**
E42/21/15	44,807	41.0 / 102.4	0.80 W	Fits, but 2.1× the volume and more core loss

I took the core data from the Ferroxcube datasheets through search results, because their site blocks direct downloads: A_min = 83, 100 and 175 mm², V_e = 6180, 8070 and 17,300 mm³. For the window I used the minimum dimensions from each datasheet drawing. Your E35 window of 181.25 mm² matches exactly.

What did I correct?

- **B_pk is 40 mT, not 45 mT.** Your handwritten line plugs in V_in·D = 36 × 0.5 = 18 V, which gives 45 mT, but you wrote ≈ 40. With $N_2 = 6$ the real duty ratio is 0.444, so V_in·D = 16 V and B_pk = 40 mT. Your typed line (with 16) was right. V_in·D stays at 16 V across the whole 36–72 V range, so the flux, and therefore the core loss, is the same at every input voltage.
- **The core loss is 0.55 W, not 0.807 W.** You read the chart at about 50 mT (B_max) and got 100 kW/m³. The design actually runs at 40 mT, where I read about 68 kW/m³, so P_core = $68 \times 10^3 \times 8.07 \times 10^{-6} \approx 0.55$ W. At 50 mT the curve is closer to 115 kW/m³ (0.93 W), and I mention that as a worst-case bound in the write-up. The chart image I made redraws the red markers at 40 mT.
- **I added a note on the chart.** The chart assumes symmetric sinusoidal flux, while a forward converter swings only from 0 up to B_pk. Reading the chart at the full 40 mT therefore overestimates the loss, which keeps the estimate on the safe side.

One assumption that could change the answer. You treated "Load 1" and "Load 2" as two operating conditions and designed for 240 W. If they're two outputs running at the same time (one 12 V winding and one 5 V winding), the primary carries both loads' reflected current, and the window needs a fourth winding. My rough count with a 3-turn 5 V winding comes to about

79 mm² of copper against 72.5 mm² usable, so the E35 would no longer fit and the E42 would win. If your instructor meant two outputs, tell me and I'll redo it for that case.

Before you compile: upload `E_core_dimensions.png` and `3F36_core_loss_40mT.png` to an Overleaf folder named `P3_Transformer/`, or change the two image paths in the file. The fragment only uses packages already in your template.

11/7/2026, 10:00:22 PM

3-new.pdf

made some changes. Now I am to have two secondary winding for two load conditions which will be operating simultaneously here. instead of notation with 1,2,3 use Np,Nsa,Nsb,Nr etc.

provide similar calculations follwing the pdf here.

11/8/2026, 6:41:44 PM

Checking where the tables and figures fall across the pages.

- Ran a command
- Read a file
- Read a file
- Read a file
- Comparing core sizes against the primary turns.
- Ran a command
- Checking the core's power loss at the chosen flux level.
- Ran a command
- Read a file
- Working out the peak flux density across both outputs.
- Ran a command
- Created a file
- Checking the secondary currents against the shared input.
- Ran a command
- Ran a command
- Read a file
- Checking where the tables and figures fall across the pages.
- Shared files
- Done

I wrote the two-output version of Problem 3 in `problem3_two_outputs.tex`, using Np, Nsa, Nsb and Nr throughout. It confirms your design: **E42/21/15 in 3F36, Np = Nr = 6, Nsa = 12, Nsb = 5**. I rechecked every number in your PDF in Python and they're all correct: 48.33 A peak, 19.73 / 8.16 / 4.08 / 1.139 A rms, 8/3/2/1 strips, 91.0 mm² of copper (35.5% fill) and B_pk = 11.4 mT. The fragment compiles in your template with no errors and no overfull lines, and it fills about five and a half pages.

What does it follow from your PDF? The same order you used: volt-second balance, volts per turn (Nsa = 12 and Nsb = 5 are the smallest whole numbers in an exact 12 : 5 ratio, which

gives 1 V/turn and D = Np/V_in), the reset limit (Np ≤ 17), the winding currents in terms of D, B_pk = V_oa·T_s/(N_sa·A_c), your winding table, the window fill and the final design table.

What did I add? The problem asks you to choose among three cores, and your PDF goes straight to the E42. I added two parts for that:

1. **The area product requirement for two outputs:** A_p ≥ [V_in·D(I_p + I_r) + V_oa·I_sa + V_ob·I_sb] / (k_w·J·B_max·f_s). At D = 0.5 this gives **A_p ≥ 14,065 mm⁴**. The E32 (12,271 mm⁴) fails; the E35 and E42 pass.
2. **Why Np = 6 is the right choice, not just a convenient one.** With Nsa and Nsb fixed, raising Np raises D, and with it the ampere-turns of every winding, so the copper needed grows with Np. Every core stays far below 50 mT, so the window is what decides the core. I tabulated the copper for each Np:

Np	D at 36 V	Copper (mm²)	Fits E35 (72.5)?	Fits E42 (102.4)?
2	0.056	51.9	Yes	Yes
3–5	0.083–0.139	74.6–82.8	No	Yes
6	**0.167**	**91.0**	No	**Yes**
7	0.194	103.7	No	No

The E35 only works with Np = 2. That means a 56 ns on-time at 72 V and a 145 A primary peak at 36 V, which isn't practical. On the E42, Np = 6 is the largest value that fits, so it gives the highest duty ratio and the lowest peak current. Your choice is the best one available under this scheme.

Core loss: ≈ 0.05 W, with a chart-based upper bound of 0.22 W. At 11.4 mT the design point sits to the left of where the 500 kHz curve starts (about 20 mT, 13 kW/m³). So 13 × 10³ × 17.3 × 10⁻⁶ = 0.22 W is a firm upper bound. Extrapolating along the curve's slope gives about 2.8 kW/m³, or 0.05 W. Both are far below 1.5 W, and the loss is the same at every input voltage because V_in·D = 6 V throughout. The new chart image marks 11.4 mT and the curve's lowest point.

Two things worth knowing about this design:

- **The core is barely used magnetically.** B_pk is 11.4 mT against an allowed 50 mT, and D is only 0.167 at 36 V. That's the cost of insisting on an exact 12 : 5 turns ratio: it forces 1 V/turn. It also means a 48 A primary peak, which is hard on the switch.
- **The exact ratio ignores the rectifier diode drops.** With about 0.5 V across each diode, the winding voltages would really need a ratio of 12.5 : 5.5, so the 5 V output will come out

slightly off. This is a common simplification for homework, but it's worth a sentence if your instructor cares about cross-regulation.

Before you compile: upload `E_core_dimensions.png` and `3F36_core_loss_11mT.png` to `P3_Transformer/` on Overleaf. No preamble changes are needed.

11/8/2026, 6:45:36 PM

Have the numerical calculations right next to each equation's derivations

11/8/2026, 6:53:43 PM

Reordering the design steps so each formula sits beside its values.

- Reordering the design steps so each formula sits beside its values.
- Ran a command
- Done

Replacing (b)+(c) with one section where each formula is followed by its numbers, then the flux block and the design-section currents.

- Ran a command
- Checking the derivation of the output voltage formula.
- Ran a command
- Read a file
- Shared files
- Done

I've updated `problem3_two_outputs.tex` so every formula now has its numbers on the same line (formula = substitution = result) or on the line right after it. It still compiles with no errors and no overfull lines.

What changed in each section?

- **(b) Winding currents and area product.** I merged the old (b) and (c) into one section. Previously the current formulas were derived first and evaluated a section later. Now each current is evaluated as soon as it's derived, at the area-product design point (V_{in} = 36 V, D = 0.5), for example $I_{p,pk} = P_o/(V_{in} \cdot D) = 290/(36 \times 0.5) = 16.11$ A. The A_p formula is followed directly by its substitution, giving 14,065 mm⁴.
- **(c) Candidate cores.** $B_{pk} = V_{oa} \cdot T_s/(N_{sa} \cdot A_c)$ now simplifies to $2 \times 10^{-6}/A_c$ and is evaluated straight away for all three cores (24.1, 20.0 and 11.4 mT). The window formula beside the core drawing now carries the E42 numbers: A_w = 256.0 mm² and A_p = 44,807 mm⁴.
- **(d) Design on the E42.** Each current line now shows the general formula before the numbers, for example $I_{sa,rms} = I_{oa} \cdot \sqrt{D} = 20 \cdot \sqrt{0.1667} = 8.16$ A. I added the strip count for each winding ($\lceil 19.73/2.73 \rceil = 8$, and so on) right below the $n = \lceil I_{rms}/2.73 \rceil$ rule. Before, those counts only appeared in the table.

- **(e) Core loss.** The extrapolation line now shows the full scaling, P_v(25 mT)·(11.4/25)^2.57, together with its numbers.

Because of the merge, the section letters now run (a) to (e). The tables, images and final results are unchanged, so the same two images in `P3_Transformer/` still apply.

11/8/2026, 6:54:47 PM

Closest to 50 mT: the E35/18/10 with N_p : N_sa : N_sb : N_r = 7 : 5 : 2 : 7. That gives B = 48.0 mT. The voltage formula then puts the 5 V output at 4.80 V, not 5.00 V. At this flux an exact 5.00 V would need 2.08 secondary turns, and a winding can't have a fraction of a turn.

Step 1: Secondary turns that put B close to 50 mT

From Faraday's law, $B_{pk} = V_{oa}\cdot T_s/(N_{sa}\cdot A_c)$, so

$$N_{sa}=\frac{V_{oa}T_s}{B_{max}A_c}$$

Core	N_sa calculated	N_sa (round up so B ≤ 50 mT)	B_pk
E35/18/10	12×2µs / (0.05×100 mm²) = 4.80	5	48.0 mT
E42/21/15	12×2µs / (0.05×175 mm²) = 2.74	3	45.7 mT

Step 2: The 5 V winding from the voltage formula

Every winding has the same volts per turn, so $V_{ob} = V_{oa}\cdot N_{sb}/N_{sa}$:

Core	Volts per turn	N_sb = N_sa×5/12	N_sb	V_ob = 12·N_sb/N_sa
E35	12/5 = 2.4 V	2.08	2	4.80 V (−4%)
E42	12/3 = 4 V	1.25	1	4.00 V (−20%)

The E42 also fails on core loss at 45.7 mT (1.56 W at 100 °C, 1.9 W at 25 °C), so the E35 is the design.

Step 3: Primary turns from the reset limit

$$D=\frac{N_pV_{oa}}{N_{sa}V_{in,min}}<0.5\;\Rightarrow\;N_p<\frac{0.5\times36\times5}{12}=7.5\;\Rightarrow\;N_p=N_r=7$$

$$D=\frac{7\times12}{5\times36}=0.467\ (36\ \text{V}),\qquad 0.233\ (72\ \text{V})$$

Step 4: Check the flux with Faraday's law

$$B_{pk}=\frac{V_{in}D\,T_s}{N_pA_c}=\frac{36\times0.467\times2\times10^{-6}}{7\times100\times10^{-6}}=48.0\ \text{mT}\;\le\;50\ \text{mT}\;\checkmark$$

Step 5: Currents, strips and window fill

Winding	Turns	I_rms (A)	Strips (I_rms ÷ 2.73 A, rounded up)	J (A/mm²)
N_p (17.14 A peak)	7	11.71	5	2.57
N_sa	5	13.66	6	2.50
N_sb	2	6.83	3	2.50
N_r	7	0.68	1	0.74

$$(7\cdot5+5\cdot6+2\cdot3+7\cdot1)\times0.91=78\times0.91=71.0\ \text{mm}^2\;\le\;0.4\times181.3=72.5\ \text{mm}^2\;\checkmark$$

The fill is 39.2% against the 40% limit. It fits, but only just.

Step 6: Core loss from the 3F36 graph

$$P_v=12\left(\frac{48.0}{20}\right)^{2.44}=101\ \text{kW/m}^3,\qquad P_{core}=101\times8.07\times10^{-3}=0.82\ \text{W}\ (100\ ^\circ\text{C})$$

$$P_{core,max}=0.82\times\frac{110}{90}=\mathbf{1.00\ W}\;\le\;1.5\ \text{W}\;\checkmark$$

==========
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I want you to use this math in the following answer script to modify the maths while following the same template:

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\problem{Problem 3Final} \label{sec:Problem3}
\problem{Problem 3} \label{sec:Problem3}
\noindent
I am to design the power transformer of a forward converter with two secondary windings that supply both loads at the same time:
\[V_{in} = 36 - 72\,\text{V}; \quad f_s = 500\,\text{kHz} \; (T_s = 2\,\mu\text{s}) \]
\[\text{Output a: } V_{oa} = 12\,\text{V}, \; I_{oa} = 20\,\text{A}; \quad \text{Output b: } V_{ob} = 5\,\text{V}, \; I_{ob} = 10\,\text{A}; \quad P_o = 12 \times 20 + 5 \times 10 = 290\,\text{W} \]
\[B_{\max} = 0.05\,\text{T}; \quad J = 3\,\text{A/mm}^2; \quad k_w = 0.4; \quad \text{Core material: 3F36}; \quad P_{core} \le 1.5\,\text{W} \]
\noindent
The windings are the primary N_p, the reset winding $N_r = N_p$, and the secondaries N_{sa} (12\,V) and N_{sb} (5\,V). I neglected the output-inductor ripple and the magnetizing current, took the reset-winding peak current as 10\% of the peak primary current, and used $A_c = A_{\min}$.

\needspace{3\baselineskip}
\subsection*{(a) Turns Relations}
\noindent
Each secondary feeds its own output inductor, and in steady state each inductor's volt-seconds balance.
\begin{gather*}
\underline{\underline{\text{Output inductor volt-second balance}}}*
\quad \left(\frac{N_s}{N_p} V_{in} - V_o \right) D T_s = V_o (1 - D) T_s\\
\implies V_o = \frac{N_s}{N_p} V_{in} D\\[1ex]
\underline{\underline{\text{Both outputs see the same } V_{in} \text{ and } D}}*
\quad \frac{V_{oa}}{N_{sa}} = \frac{V_{ob}}{N_{sb}} = \frac{V_{in} D}{N_p} \quad \text{(volts per turn)}\\
\implies \frac{N_{sa}}{N_{sb}} = \frac{V_{oa}}{V_{ob}} = \frac{12}{5}\\
\implies \boxed{N_{sa} = 12, \; N_{sb} = 5} \quad \text{(smallest whole numbers with the exact ratio)}\\
\implies \frac{V_{in} D}{N_p} = \frac{12}{12} = 1\,\text{V/turn}\\
\implies D = \frac{N_p (1\,\text{V})}{V_{in}}\\[1ex]
\underline{\underline{\text{Reset limit}}}*
\quad D + \frac{N_r}{N_p} D \le 1\\
\implies D_{\max} = 0.5 \quad (N_r = N_p)\\
\implies \frac{N_p \times 1}{36} < 0.5 \quad \text{at } V_{in,\min} = 36\,\text{V}\\
\implies N_p \le 17
\end{gather*}

\needspace{3\baselineskip}
\subsection*{(b) Winding Currents and Required Area Product}

```
\noindent
The area product is sized at the worst case allowed by the reset limit, $V_{in} = 36\,\text{V}$ and $D = D_{\max} = 0.5$ ($V_{in} D = 18\,\text{V}$). Each current is derived and evaluated at that point.
\begin{gather*}
\underline{\underline{\text{Primary}}}\\*
\quad N_p I_{p,pk} = N_{sa} I_{oa} + N_{sb} I_{ob}\\
\implies I_{p,pk} = \frac{V_{oa} I_{oa} + V_{ob} I_{ob}}{V_{in} D} = \frac{P_o}{V_{in} D} = \frac{290}{36 \times 0.5} = 16.11\,\text{A}\\
\implies I_{p,rms} = I_{p,pk} \sqrt{D} = \frac{P_o}{V_{in} \sqrt{D}} = 16.11 \sqrt{0.5} = 11.39\,\text{A}\\[1ex]
\underline{\underline{\text{Secondaries}}}\\*
\quad I_{sa,rms} = I_{oa} \sqrt{D} = 20 \sqrt{0.5} = 14.14\,\text{A}\\
\implies I_{sb,rms} = I_{ob} \sqrt{D} = 10 \sqrt{0.5} = 7.07\,\text{A}\\[1ex]
\underline{\underline{\text{Reset}}}\\*
\quad I_{r,pk} = 0.1\, I_{p,pk} = 0.1 \times 16.11 = 1.611\,\text{A}\\
\implies I_{r,rms} = I_{r,pk} \sqrt{\frac{D}{3}} = 1.611 \sqrt{\frac{0.5}{3}} = 0.658\,\text{A}\\[1ex]


\noindent
\begin{minipage}[c]{0.56\textwidth}
   \begin{gather*}
   \underline{\underline{\text{Window of a core set}}}\\*
   \quad A_w = \frac{B - C}{2} \times 2E\\
   \implies A_w = (B - C)\, E\\
   % \implies A_p = A_{\min} A_w\\[1ex]
   \text{E42/21/15:}\\*
   \quad A_w = (29.5 - 12.2) \times 14.8 = 256.0\,\text{mm}^2\\
   k_w A_w = 0.4 \times 256.0 = 102.4\,\text{mm}^2 \quad
   % \implies A_p = 175 \times 256.0 = 44\,807\,\text{mm}^4
   \end{gather*}
\end{minipage}
\vspace{4pt}

% \noindent
% \underline{\underline{\text{Faraday's law and window}}}\\*
% \quad N_p = \frac{V_{in} D}{B_{\max} A_c f_s}\\
% % \implies \frac{N_p I_{p,rms} + N_{sa} I_{sa,rms} + N_{sb} I_{sb,rms} + N_r I_{r,rms}}{J} \le k_w A_w\\
% % \implies A_p = A_c A_w \ge \frac{V_{in} D \left( I_{p,rms} + I_{r,rms} \right) + V_{oa} I_{sa,rms} + V_{ob} I_{sb,rms}}{k_w J B_{\max} f_s}\\
% % \implies A_p \ge \frac{18 \times (11.39 + 0.658) + 12 \times 14.14 + 5 \times 7.07}{0.4 \times 3 \times 10^6 \times 0.05 \times 500 \times 10^3} = \frac{421.96}{3 \times 10^{10}}\\
% % \implies \boxed{A_p \ge 1.4065 \times 10^{-8}\,\text{m}^4 = 14\,065\,\text{mm}^4}
% \end{gather*}
\begin{gather*}
\underline{\underline{\text{Faraday's law and window}}}\\
N_p = \frac{V_{in} D}{B_{\max} A_c f_s}\\
\implies \quad N_p A_c B_{pk} = V_{in} D T_s = N_p \frac{V_{oa}}{N_{sa}} T_s\\
\implies B_{pk} = \frac{V_{oa} T_s}{N_{sa} A_c} = \frac{12 \times 2 \times 10^{-6}}{12 \times A_c} = \frac{2 \times 10^{-6}}{A_c}\\
% \implies B_{pk} = \frac{2 \times 10^{-6}}{83 \times 10^{-6}} =
```

```
24.1\,\text{mT} \quad \text{(E32/16/9)}\\
% \implies B_{pk} = \frac{2 \times 10^{-6}}{100 \times 10^{-6}} =
20.0\,\text{mT} \quad \text{(E35/18/10)}\\
% \implies B_{pk} = \frac{2 \times 10^{-6}}{175 \times 10^{-6}} =
11.4\,\text{mT} \quad \text{(E42/21/15)}
\end{gather*}

\begin{gather*}
\boxed{\text{Choose} \quad \text{E42/21/15, 3F36}; \quad N_p = N_r = 6; \quad
N_{sa} = 12; \quad N_{sb} = 5}
\end{gather*}

% \begin{table}[!htbp]
% \centering
% \caption{Candidate cores (minimum dimensions from the Ferroxcube datasheet
drawings; required $A_p \ge 14\,065\,\text{mm}^4$)}
% \label{tab:P3 two output cores}
% \setlength{\tabcolsep}{4pt}
% \begin{tabular}{@{}lccccccccc@{}}
% \toprule
% Core & $A_{\min}$ (mm$^2$) & $B$ & $C$ & $E$ (mm) & $A_w$ (mm$^2$) &
$A_p$ (mm$^4$) & $A_p$ OK? & $B_{pk}$ (mT) & $k_w A_w$ (mm$^2$)\\
% \midrule
% E32/16/9  & 83  & 22.7 & 9.5  & 11.2 & 147.8 & 12\,271 & No  & 24.1 & 59.1\\
% E35/18/10 & 100 & 24.5 & 10.0 & 12.5 & 181.3 & 18\,125 & Yes & 20.0 & 72.5\\
% E42/21/15 & 175 & 29.5 & 12.2 & 14.8 & 256.0 & 44\,807 & Yes & 11.4 & 102.4\\
% \bottomrule
% \end{tabular}
% \end{table}
\noindent
I will use 13\,mm $\times$ 0.07\,mm strips, $A_{st} = 0.91\,\text{mm}^2$.


\noindent

\begin{gather*}
\underline{\underline{\text{Duty ratio}}}\\*
\quad D = \frac{N_p \times 1\,\text{V}}{V_{in}} = \frac{6}{36} = 0.167 \quad
(36\,\text{V}) \qquad\qquad D = \frac{6}{72} = 0.083 \quad (72\,\text{V})\\[1ex]
\underline{\underline{\text{Peak flux density}}}\\*
\quad B_{pk} = \frac{V_{oa} T_s}{N_{sa} A_c} = \frac{12 \times 2 \times
10^{-6}}{12 \times 175 \times 10^{-6}}\\
\implies \boxed{B_{pk} = 11.4\,\text{mT} \le 50\,\text{mT}} \quad \text{(Valid)}\\
[1ex]
\underline{\underline{\text{Winding currents at } V_{in} = 36\,\text{V}}}\\*
\quad I_{p,pk} = \frac{N_{sa} I_{oa} + N_{sb} I_{ob}}{N_p} = \frac{12 \times 20
+ 5 \times 10}{6} = 48.33\,\text{A}\\
\implies I_{p,rms} = I_{p,pk} \sqrt{D} = 48.33 \sqrt{0.1667} = 19.73\,\text{A}\\
\implies I_{sa,rms} = I_{oa} \sqrt{D} = 20 \sqrt{0.1667} = 8.16\,\text{A}\\
\implies I_{sb,rms} = I_{ob} \sqrt{D} = 10 \sqrt{0.1667} = 4.08\,\text{A}\\
\implies I_{r,pk} = 0.1\, I_{p,pk} = 0.1 \times 48.33 = 4.833\,\text{A}\\
\implies I_{r,rms} = I_{r,pk} \sqrt{\frac{D}{3}} = 4.833 \sqrt{\frac{0.1667}{3}}
= 1.139\,\text{A}\\
\implies N_p I_{p,pk} = 6 \times 48.33 = 290 = 12 \times 20 + 5 \times 10 \quad
\text{(ampere-turn check)}
```

```
\end{gather*}
\noindent
Each winding uses as many parallel strips as it needs to carry its RMS current at $J
\le 3\,\text{A/mm}^2$.
\begin{gather*}
\underline{\underline{\text{Strips per turn}}}\\*
\quad I_{st} = J A_{st} = 3 \times 0.91 = 2.73\,\text{A per strip}\\
\implies n = \left\lceil \frac{I_{rms}}{2.73} \right\rceil\\
\implies n_p = \left\lceil \frac{19.73}{2.73} \right\rceil = \lceil 7.23 \rceil = 8\\
\implies n_{sa} = \left\lceil \frac{8.16}{2.73} \right\rceil = \lceil 2.99 \rceil = 3\\
\implies n_{sb} = \left\lceil \frac{4.08}{2.73} \right\rceil = \lceil 1.50 \rceil = 2\\
\implies n_r = \left\lceil \frac{1.139}{2.73} \right\rceil = \lceil 0.42 \rceil = 1
\end{gather*}
\begin{table}[!htbp]
\centering
\caption{Winding table for the E42/21/15 design}
\label{tab:P3 winding table}
\begin{tabular}{@{}lccccc@{}}
\toprule
Winding & Turns & $I_{rms}$ (A) & $I_{rms}/(J A_{st})$ & $n$ & Copper (mm$^2$)
& $J$ (A/mm$^2$)\\
\midrule
$N_p$    & 6  & 19.73 & $19.73/2.73 = 7.23$ & 8 & 7.28 & 2.71\\
$N_{sa}$ & 12 & 8.16  & $8.16/2.73 = 2.99$  & 3 & 2.73 & 2.99\\
$N_{sb}$ & 5  & 4.08  & $4.08/2.73 = 1.50$  & 2 & 1.82 & 2.24\\
$N_r$    & 6  & 1.139 & $1.139/2.73 = 0.42$ & 1 & 0.91 & 1.25\\
\bottomrule
\end{tabular}
\end{table}
\begin{gather*}
\underline{\underline{\text{Window fill}}}\\*
\quad A_{cu} = (N_p n_p + N_{sa} n_{sa} + N_{sb} n_{sb} + N_r n_r) A_{st}\\
\implies A_{cu} = (6 \times 8 + 12 \times 3 + 5 \times 2 + 6 \times 1) \times 0.91 =
100 \times 0.91\\
\implies \boxed{A_{cu} = 91.0\,\text{mm}^2 \le k_w A_w = 0.4 \times 256.0 =
102.4\,\text{mm}^2} \quad \text{(Valid)}\\
\end{gather*}
\noindent



\needspace{3\baselineskip}
\subsection*{(e) Maximum Core Loss}
\noindent
Since $V_{in} D = N_p \times 1\,\text{V} = 6\,\text{V}$ at every input voltage,
$B_{pk} = 11.4\,\text{mT}$ over the whole input range, and the core loss is the
same at $36\,\text{V}$ and $72\,\text{V}$.
% \begin{figure}[!htbp]
%     \centering
%     \bigfig[0.34]{P3_Transformer/3F36_core_loss_11mT.png}
%     \caption{3F36 specific power loss $P_v$ vs. peak flux density $\hat{B}$ at $T
= 100\,^\circ\text{C}$. The design point ($\hat{B} = 11.4\,\text{mT}$) lies left of
the lowest point of the $500\,\text{kHz}$ curve ($20\,\text{mT}$, $\approx
13\,\text{kW/m}^3$).}
%     \label{fig:P3 3F36 core loss 11mT}
% \end{figure}
```

```
\begin{gather*}
\underline{\underline{\text{Upper bound from the chart}}}\\*
\quad P_v (11.4\,\text{mT}) < P_v (20\,\text{mT}) \approx 13\,\text{kW/m}^3
\quad \text{(Fig.~\ref{fig:P3 3F36 core loss 11mT})}\\
\implies P_{core} < 13 \times 10^3 \times 17.3 \times 10^{-6}\\
\implies P_{core} < 0.22\,\text{W}\\[1ex]
\underline{\underline{\text{Extrapolated along the curve slope}}}\\*
\quad P_v (11.4\,\text{mT}) = P_v (25\,\text{mT}) \left( \frac{11.4}{25}
\right)^{2.57} \approx 20.7 \times \left( \frac{11.4}{25} \right)^{2.57} \approx
2.8\,\text{kW/m}^3\\
\implies P_{core} \approx 2.8 \times 10^3 \times 17.3 \times 10^{-6}\\
\implies \boxed{P_{core,\max} \approx 0.05\,\text{W} \; (< 0.22\,\text{W}) \le
1.5\,\text{W}} \quad \text{(Valid)}
\end{gather*}

\begin{table}[!htbp]
\centering
\caption{Final transformer design}
\label{tab:P3 two output final}
\begin{tabular}{@{}ll@{}}
\toprule
Item & Value\\
\midrule
Core & E42/21/15 pair, 3F36 ($A_c = 175\,\text{mm}^2$, $A_w =
256.0\,\text{mm}^2$, $V_e = 17\,300\,\text{mm}^3$)\\
Primary $N_p$ & 6 turns $\times$ 8 strips (7.28\,mm$^2$, 19.73\,A rms, 48.3\,A
peak)\\
Secondary $N_{sa}$ (12\,V, 20\,A) & 12 turns $\times$ 3 strips (2.73\,mm$^2$,
8.16\,A rms)\\
Secondary $N_{sb}$ (5\,V, 10\,A) & 5 turns $\times$ 2 strips (1.82\,mm$^2$,
4.08\,A rms)\\
Reset $N_r$ & 6 turns $\times$ 1 strip (0.91\,mm$^2$, 1.139\,A rms)\\
Turns ratio $N_p : N_{sa} : N_{sb} : N_r$ & 6 : 12 : 5 : 6\\
Duty ratio & 0.167 at 36\,V, 0.083 at 72\,V\\
Peak flux density & 11.4\,mT ($\le 50\,\text{mT}$)\\
Window fill & 91.0\,mm$^2$ of 102.4\,mm$^2$ usable (35.5\%)\\
Core loss & $\approx 0.05\,\text{W}$, below $0.22\,\text{W}$ ($\le
1.5\,\text{W}$)\\
\bottomrule
\end{tabular}
\end{table}



\noindent
```

11/8/2026, 8:34:24 PM

- Restructuring the turns and core-selection sections.
- Searched the web
- Testing a fitted loss formula against the table.
- Ran a command

Created a file

Ran a command

Read a file

Shared files

Done

I rewrote your script with the new math in `problem3_final.tex`. The final design is the **E35/18/10 with Np : Nsa : Nsb : Nr = 7 : 5 : 2 : 7**. It keeps the same template: the same intro, `gather*` blocks with each formula and its numbers together, the window drawing beside the equations, the strip and winding tables, and the final design table. It compiles with no errors and no overfull lines, and it fills about four and a half pages.

I rechecked every number from your notes in Python, and all of them hold up: B = 48.0 mT, Vob = 4.80 V, D = 0.467 and 0.233, Ip,pk = 17.14 A, the RMS currents 11.71, 13.66, 6.83 and 0.676 A, 5/6/3/1 strips, 71.0 mm² of copper (39.2% fill), and a core loss of 0.82 W rising to 1.00 W.

What did I add or change?

- **The E32 is now part of the selection.** Your notes only compared the E35 and E42, but the problem asks you to choose among all three. The E32 gives Nsa = 6 (48.2 mT), but Nsb = 2.5, so the 5 V output comes out at 4.0 V or 6.0 V (20% off either way). Its window is also too small, at 76.4 mm² of copper against 59.1 mm² usable. All three cores go through Steps 1–3 and into one comparison table.
- **The 110/90 temperature factor now has a source.** The 3F36 datasheet lists ≈110 kW/m³ at 25 °C and ≈90 kW/m³ at 100 °C (500 kHz, 50 mT), and the core-loss section now says so. Without that, the 1.00 W worst case had nothing to back it up.
- **Nsa needs 6 strips, not 5, and only just.** 13.66/2.73 = 5.005, and with 5 strips J would be 3.003 A/mm², just over the limit. I showed ⌈5.005⌉ = 6 explicitly so a grader doesn't wonder about it.
- **I cleaned up the script.** Your version had `\problem` twice, a `Fig.~\ref` pointing at a figure you'd commented out (it would print "??"), and commented-out E42 blocks that no longer apply. I removed all of those. The section letters now run (a) to (c), and the old D = 0.5 area-product calculation is gone, since the new method sizes the turns directly from B_max.

Two things to know about the design:

- **The 5 V output really is 4.80 V.** That's the closest any of the three cores gets with whole turns, and the write-up says so plainly.
- **The window margin is small.** It's 71.0 against 72.5 mm², so any extra insulation layer could push it over.

The only image it needs is `E_core_dimensions.png` in `P3_Transformer/`.

11/8/2026, 8:36:51 PM